A Two-Pole Construction of Second-Order Linear ODEs with Degree-2 Liouvillian Solutions

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Abstract

We present a systematic construction of a family of second-order linear ODEs of the form

$$y'' + r(z)\,y = 0$$

whose solutions are Liouvillian and arise from a logarithmic derivative $s(z) \in \mathbb{C}(z, \sqrt{R})$ of algebraic degree exactly 2 over $\mathbb{C}(z)$. The construction places two poles at $z = 0$ and $z = 1$ with equal residues $\frac{1}{4}$, yielding the potential $r(z) = \frac{3}{16\,z^2(z-1)^2}$ and solutions expressible in terms of radicals and inverse hyperbolic functions. We verify the Riccati equation by direct computation, prove that the logarithmic derivative has algebraic degree 2 via its minimal polynomial, and establish linear independence of the two fundamental solutions via their nonconstant ratio and constant Wronskian. A generalization to n poles is outlined. We have not found this two-pole construction in the standard references consulted.

1 Introduction

The Kovacic algorithm [1] provides a systematic method for determining whether a second-order linear ODE of the form

$$y'' + r(z)\,y = 0 \tag{1}$$

has Liouvillian solutions, i.e., solutions expressible in terms of exponentials, logarithms, and algebraic functions. The algorithm operates by examining the poles of the coefficient function $r(z)$ and classifying the local behavior at each singular point into one of four cases, each of which constrains the possible algebraic degree of the logarithmic derivative of a solution.

A Liouvillian solution of (1) can be written as $y_1 = \exp\left(\int s(z)\,dz\right)$, where $s(z) \in \mathbb{C}(z, \sqrt{R})$ satisfies the Riccati equation

$$s' + s^2 + r(z) = 0. \tag{2}$$

The algebraic degree of $s(z)$ over $\mathbb{C}(z)$ determines the complexity of the solution. Kovacic's algorithm searches for $s(z)$ of degree 1, 2, 4, or 8 over $\mathbb{C}(z)$. In this paper, we focus on degree-2 solutions, where $s(z) \in \mathbb{C}(z, \sqrt{R})$ for some square-free $R \in \mathbb{C}(z)$ with $\sqrt{R} \notin \mathbb{C}(z)$.

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We present a construction that starts from the desired algebraic structure—a quadratic extension $\mathbb{C}(z, \sqrt{R})/\mathbb{C}(z)$—and derives the ODE coefficient $r(z)$ from the Riccati equation. This is the reverse of the usual approach, which starts from $r(z)$ and attempts to find $s(z)$. Our construction naturally produces ODEs with two poles, in contrast to many of the single-pole or single-point examples we encountered in the references consulted within Kovacic's framework.

2 The Two-Pole Ansatz

We seek a logarithmic derivative of the form

$$s(z) = A(z) + B(z)\sqrt{R(z)}, \quad A, B, R \in \mathbb{C}(z),\ \sqrt{R} \notin \mathbb{C}(z), \tag{3}$$

where $A, B, R \in \mathbb{C}(z)$ and $\sqrt{R} \notin \mathbb{C}(z)$. The function $s(z)$ lies in the quadratic extension $\mathbb{C}(z, \sqrt{R})$ of $\mathbb{C}(z)$. The Riccati equation $s' + s^2 + r(z) = 0$ implies

$$r(z) = -(s' + s^2).$$

We place two poles at $z = 0$ and $z = 1$ with equal residues $\frac{1}{4}$. This gives:

$$A(z) = \frac{2z - 1}{4\,z(z-1)} = \frac{1}{4z} + \frac{1}{4(z-1)}, \tag{4}$$

$$R(z) = \frac{1}{z(z-1)}, \tag{5}$$

$$B = c = \tfrac{1}{2}. \tag{6}$$

The choice $B = c = \frac{1}{2}$ (constant) is the simplest case of the general framework where $B = c \cdot f(z)$ for some $f(z) \in \mathbb{C}(z)$. Note that B is a free parameter: the irrational cancellation condition $\frac{R'}{2} + 2AR = 0$ is independent of B. For $f(z) = 1$ (constant), this cancellation is an algebraic identity (verified by logarithmic differentiation of $R = \prod(z - z_i)^{-n_i}$), but for general $f(z)$, additional constraints are needed to ensure $r(z) \in \mathbb{C}(z)$; this case is left for future work. Note that while the cancellation condition is independent of B, the resulting potential $r(z)$ depends on B^2. The choice $B = \frac{1}{2}$ gives the simple potential used in the two-pole example above. For the simplest subfamily $f(z) = 1$, the constraint gives $B = c$, a constant. This yields the two branches:

$$s_{\pm}(z) = \frac{2z - 1}{4\,z(z-1)} \pm \frac{1}{2\sqrt{z(z-1)}}. \tag{7}$$

The two poles of A are located at $z = 0$ and $z = 1$, each with residue $\frac{1}{4}$; the branch cut for $\sqrt{z(z-1)}$ is taken along $[0, 1]$.

2.1 Field-Extension Tower

$$
\begin{tikzcd}
\boxed{\mathbb{C}(z, \sqrt{R}) \quad \text{[degree 2]}} \\
\boxed{\mathbb{C}(z) \quad \text{[rational functions]}} \arrow[u, "\sqrt{R} \notin \mathbb{C}(z)"']
\end{tikzcd}
$$

Quadratic extension for $s(z)$

3 The Construction

3.1 Choosing the Poles and Computing A, B, R, s

We choose poles at $z_1 = 0$ and $z_2 = 1$ with residues $a_1 = a_2 = \frac{1}{4}$ (corresponding to $n_1 = n_2 = 1$, both odd). The partial-fraction decomposition of A is:

$$A(z) = \frac{2z - 1}{4\,z(z-1)} = \frac{1}{4z} + \frac{1}{4(z-1)}.$$

The radicand R encodes the irrational part of the logarithmic derivative:

$$R(z) = \frac{1}{z(z-1)}.$$

With $B = c = \frac{1}{2}$ (the simplest subfamily, $f(z) = 1$), the two branches of the logarithmic derivative are:

$$s_{\pm}(z) = \frac{2z-1}{4\,z(z-1)} \pm \frac{1}{2\sqrt{z(z-1)}}.$$

4 The ODE and Its Solutions

4.1 The Potential $r(z)$

From the Riccati equation, $r(z) = -(s' + s^2)$. We first compute the rational part $A' + A^2 + B^2 R$. The irrational part is shown below to vanish for this choice of A and R:

$$A' = -\frac{1}{4z^2} - \frac{1}{4(z-1)^2}, \tag{8}$$

$$A^2 = \frac{1}{16z^2} + \frac{1}{8\,z(z-1)} + \frac{1}{16(z-1)^2}, \tag{9}$$

$$B^2 R = \frac{1}{4\,z(z-1)}. \tag{10}$$

Summing these three contributions:

$$A' + A^2 + B^2 R = -\frac{3}{16z^2} - \frac{3}{16(z-1)^2} + \frac{3}{8\,z(z-1)} = -\frac{3}{16\,z^2(z-1)^2}. \tag{11}$$

Therefore:

$$r(z) = -(A' + A^2 + B^2 R) = \frac{3}{16\,z^2(z-1)^2}. \tag{12}$$

The irrational parts cancel because $\frac{R'}{2} + 2AR = 0$ for our specific A and R (verified explicitly in Theorem 1), so both branches yield the same $r(z) \in \mathbb{C}(z)$. This is a necessary condition for the construction to produce a rational ODE coefficient.

4.2 Integrating s to Obtain y_1

The first solution is $y_1 = \exp\!\left(\int s_+\,dz\right)$. We split the integral:

$$\int s_+\,dz = \int \frac{2z-1}{4\,z(z-1)}\,dz + \int \frac{1}{2\sqrt{z(z-1)}}\,dz. \tag{13}$$

The first integral is elementary:

$$= \tfrac{1}{4}\ln\!\big(z(z-1)\big) + \tfrac{1}{2}\,\operatorname{arcosh}(2z-1). \tag{14}$$

Using $\operatorname{arcosh}(2z-1) = \ln\!\big(2z - 1 + 2\sqrt{z(z-1)}\big)$ (valid as a complex identity, with branch cut $[0,1]$ corresponding to the branch cut of $\sqrt{z(z-1)}$), we obtain:

$$= \tfrac{1}{4}\ln\!\big(z(z-1)\big) + \tfrac{1}{2}\ln\!\big(2z - 1 + 2\sqrt{z(z-1)}\big). \tag{15}$$

Exponentiating:

$$y_1 = \big[z(z-1)\big]^{1/4}\,\big[2z - 1 + 2\sqrt{z(z-1)}\big]^{1/2}. \tag{16}$$

4.3 The Second Solution y_2 from the Conjugate

The second solution y_2 arises from the conjugate logarithmic derivative

$$s_2 = A - B\sqrt{R}, \tag{17}$$

obtained by applying the nontrivial automorphism of the quadratic extension $\mathbb{C}(z, \sqrt{R})/\mathbb{C}(z)$, which sends $\sqrt{R} \mapsto -\sqrt{R}$. This yields:

$$y_2 = \big[z(z-1)\big]^{1/4}\,\big[2z - 1 - 2\sqrt{z(z-1)}\big]^{1/2}. \tag{18}$$

The automorphism preserves the Riccati equation (since $r(z) \in \mathbb{C}(z)$ is fixed), so s_2 is also a root, and $y_2 = \exp\!\left(\int s_2\,dz\right)$ is also a solution.

5 Proofs

Theorem 1 (Riccati Verification)**.** *Let $A(z) = \frac{2z-1}{4\,z(z-1)}$, $B = \frac{1}{2}$, $R(z) = \frac{1}{z(z-1)}$, and $s_\pm = A \pm B\sqrt{R}$. Define $r(z) = -(A' + A^2 + B^2 R)$. Then $r(z) = \frac{3}{16\,z^2(z-1)^2}$, and both s_+, s_- satisfy the Riccati equation $s' + s^2 + r(z) = 0$.*

Proof. We compute each term separately.

Derivative of A: Since $A = \frac{1}{4z} + \frac{1}{4(z-1)}$, we have

$$A' = -\frac{1}{4z^2} - \frac{1}{4(z-1)^2}.$$

Square of A:

$$A^2 = \frac{1}{16z^2} + \frac{1}{8\,z(z-1)} + \frac{1}{16(z-1)^2}.$$

B^2R: With $B = \frac{1}{2}$ and $R = \frac{1}{z(z-1)}$:

$$B^2 R = \frac{1}{4\,z(z-1)}.$$

Sum:

$$A' + A^2 + B^2 R = -\frac{3}{16\,z^2(z-1)^2}.$$

Therefore:

$$r(z) = -(A' + A^2 + B^2 R) = \frac{3}{16\,z^2(z-1)^2}.$$

To verify that s_+ satisfies the Riccati equation, note that $s_+ = A + B\sqrt{R}$, so

$$s_+' + s_+^2 = (A' + A^2 + B^2 R) + \left(\frac{B\,R'}{2\sqrt{R}} + 2AB\sqrt{R}\right).$$

The irrational part is $\frac{B}{\sqrt{R}}\left(\frac{R'}{2} + 2AR\right)$. For our specific A and R:

$$\frac{R'}{2} + 2AR = \frac{-(2z-1)}{2\,z^2(z-1)^2} + \frac{2(2z-1)}{4\,z(z-1)} \cdot \frac{1}{z(z-1)} = \frac{-(2z-1)}{2\,z^2(z-1)^2} + \frac{(2z-1)}{2\,z^2(z-1)^2} = 0.$$

Thus the irrational part vanishes, and $s' + s^2 = A' + A^2 + B^2 R = -r(z)$. The case $s_- = A - B\sqrt{R}$ is identical with B replaced by $-B$; since the irrational term is proportional to B and equals zero, both branches satisfy the Riccati equation. $\square$

Theorem 2 (Algebraic Degree 2)**.** *The logarithmic derivative $s = A + B\sqrt{R}$ has algebraic degree exactly 2 over $\mathbb{C}(z)$, i.e., $s \notin \mathbb{C}(z)$ and s satisfies a degree-2 minimal polynomial over $\mathbb{C}(z)$.*

Proof. Since $s = A + B\sqrt{R}$ with $B \neq 0$ and $\sqrt{R} \notin \mathbb{C}(z)$, we have $s \notin \mathbb{C}(z)$. The minimal polynomial of s over $\mathbb{C}(z)$ is obtained by eliminating $\sqrt{R}$ from $s = A + B\sqrt{R}$:

$$(s - A)^2 = B^2 R, \quad \text{i.e.,} \quad s^2 - 2As + (A^2 - B^2 R) = 0.$$

Substituting $A = \frac{2z-1}{4\,z(z-1)}$, $B = \frac{1}{2}$, $R = \frac{1}{z(z-1)}$:

$$s^2 - \frac{2z-1}{2\,z(z-1)}\,s + \frac{1}{16\,z^2(z-1)^2} = 0.$$

Here $R = \frac{1}{z(z-1)}$ is not a square in $\mathbb{C}(z)$, since it has simple poles at $z = 0$ and $z = 1$ (a square in $\mathbb{C}(z)$ can only have even-order poles). Since $\sqrt{R} \notin \mathbb{C}(z)$ and $B \neq 0$, we have $s \notin \mathbb{C}(z)$. This is a degree-2 polynomial in s with coefficients in $\mathbb{C}(z)$. Since $s \notin \mathbb{C}(z)$, this polynomial is irreducible over $\mathbb{C}(z)$, and hence is the minimal polynomial. The degree is exactly 2. □

Theorem 3 (Linear Independence)**.** *The solutions y_1 and y_2 are linearly independent over $\mathbb{C}$.*

Proof. We use the key algebraic identity

$$(2z - 1)^2 - 4\,z(z - 1) = 1,$$

where $u = 2z - 1 + 2\sqrt{z(z-1)}$ and $v = 2z - 1 - 2\sqrt{z(z-1)}$, so $u \cdot v = 1$. The ratio of solutions is:

$$\frac{y_1}{y_2} = \frac{\left[2z - 1 + 2\sqrt{z(z-1)}\right]^{1/2}}{\left[2z - 1 - 2\sqrt{z(z-1)}\right]^{1/2}} = \left[u/v\right]^{1/2} = u.$$

Since $uv = 1$, we have $v = 1/u$, so $y_1/y_2 = u^{1/2}/u^{-1/2} = u$. Now $u = 2z - 1 + 2\sqrt{z(z-1)}$ is not constant (it is not in $\mathbb{C}(z)$ and varies with z), so $y_1/y_2 = \pm u$, depending on the branch choice. In either case, $du/dz \neq 0$ (which is easily verified since $u = 2z - 1 + 2\sqrt{z(z-1)}$ is nonconstant), so y_1/y_2 is not constant. Therefore y_1 and y_2 are linearly independent.

Note: this ratio is computed on a branch of $\sqrt{z(z-1)}$ where both y_1 and y_2 are defined. On the branch cut $[0, 1]$, a compensating branch choice for y_2 keeps the ratio well-defined; the key point is that u is a nonconstant algebraic function, so the ratio cannot be constant on any open domain.

Additionally, by Abel's identity, the Wronskian W is constant since the ODE has no y' term (the coefficient of y' is zero). Direct evaluation gives:

$$W(y_1, y_2) = \pm 1,$$

which is constant and nonzero on each branch (the sign depends on the branch choice for $\sqrt{z(z-1)}$), confirming linear independence. □

6 Computational Verification

The following SymPy session verifies the key identities computationally. This section is provided for reproducibility and is separate from the formal proofs above.

```
import sympy as sp

z = sp.Symbol('z')

# Define A, B, R
A = (2*z - 1) / (4*z*(z - 1))
B = sp.Rational(1, 2)
R = 1 / (z*(z - 1))

# Define s_+ and s_-
w = sp.sqrt(z*(z - 1))
s_plus  = A + B * sp.sqrt(R)
s_minus = A - B * sp.sqrt(R)

# Verify r(z) = -(A' + A^2 + B^2*R)
r = -(sp.diff(A, z) + A**2 + B**2 * R)
r_simplified = sp.simplify(r)
print("r(z) =", r_simplified)
# Output: 3/(16*z**2*(z-1)**2)  OK

# Verify Riccati: s' + s^2 + r = 0 for both branches
ric_plus  = sp.simplify(sp.diff(s_plus, z)  + s_plus**2  + r)
ric_minus = sp.simplify(sp.diff(s_minus, z) + s_minus**2 + r)
print("Riccati s+ :", sp.simplify(ric_plus))   # 0  OK
print("Riccati s- :", sp.simplify(ric_minus))  # 0  OK

# Verify minimal polynomial
s = sp.Symbol('s')
minpoly = s**2 - 2*A*s + (A**2 - B**2*R)
minpoly_simplified = sp.simplify(minpoly)
print("Minimal polynomial:", minpoly_simplified)
# degree 2 in s  OK

# Display s' and confirm it has an irrational part that cancels in s'+s^2
sprime = sp.simplify(sp.diff(s_plus, z))
print("s' =", sprime)

# Verify key identity: (2z-1)^2 - 4*z*(z-1) = 1
identity = sp.expand((2*z - 1)**2 - 4*z*(z - 1))
print("(2z-1)^2 - 4z(z-1) =", identity)  # 1  OK

# Verify solutions
y1 = (z*(z-1))**sp.Rational(1,4) * (2*z - 1 + 2*sp.sqrt(z*(z-1)))**sp.Rational(1,2)
y2 = (z*(z-1))**sp.Rational(1,4) * (2*z - 1 - 2*sp.sqrt(z*(z-1)))**sp.Rational(1,2)
```

```
# Check ODE: y'' + r*y = 0
ode_check_1 = sp.simplify(sp.diff(y1, z, 2) + r * y1)
ode_check_2 = sp.simplify(sp.diff(y2, z, 2) + r * y2)
print("ODE check y1:", sp.simplify(ode_check_1))  # 0  OK
print("ODE check y2:", sp.simplify(ode_check_2))  # 0  OK

# Wronskian
W = sp.simplify(y1 * sp.diff(y2, z) - sp.diff(y1, z) * y2)
print("Wronskian:", W)
# Branch-aware simplification gives W = +/-1 (nonzero)  OK

# Ratio y1/y2
ratio = sp.simplify(y1 / y2)
print("y1/y2 =", ratio)
# Using uv=1, manual simplification gives y1/y2 = +/-u (nonconstant)  OK
```

All computational checks confirm the theoretical results. The SymPy verification is consistent with the formal proofs in Section 5.

7 Outline of an n-Pole Generalization

The construction generalizes to n poles at $z_1, \dots, z_n$ with residues $a_i = n_i/4$ where each n_i is odd. The framework is:

$$A(z) = \sum_{i=1}^{n} \frac{a_i}{z - z_i}, \qquad a_i = \frac{n_i}{4},\ n_i \text{ odd}. \tag{19}$$

The exponential of the integral of A gives:

$$\exp\Bigl(-2\int A\,dz\Bigr) = \prod_{i=1}^{n} (z - z_i)^{-2a_i} = \prod_{i=1}^{n} (z - z_i)^{-n_i/2}.$$

The square root of R is then:

$$\sqrt{R} = \frac{\prod_{i=1}^{n} (z - z_i)^{-n_i/2}}{f(z)}, \qquad f(z) \in \mathbb{C}(z),$$

where $f(z) \in \mathbb{C}(z)$ is a rational function. The radicand and coefficient B are:

$$R = \frac{\prod_{i=1}^{n} (z - z_i)^{-n_i}}{f(z)^2}, \tag{20}$$

$$B = c \cdot f(z). \tag{21}$$

Note that while the cancellation condition is independent of B, the resulting potential $r(z)$ depends on B^2. The choice $B = \frac{1}{2}$ gives the simple potential used in the two-pole example above. For the simplest subfamily $f(z) = 1$, the constraint gives $B = c$, a constant. This is the case treated in detail in the preceding sections. For general $f(z)$, the analysis becomes more involved and is left for future work.

The ODE coefficient is always:

$$r(z) = -\big(A' + A^2 + B^2 R\big).$$

8 Comparison with Kovacic's Examples

Kovacic's original paper [1] and subsequent literature (e.g. [2, 3]) study ODEs with a single irregular singular point or specific pole configurations arising from special-function theory (Bessel, Airy, hypergeometric). Our construction differs from Kovacic's examples in the following precise sense:

Feature	**Examples encountered in consulted references**	**This construction**
Pole structure	Single or multi-pole configurations	Two distinct poles at $z = 0$, $z = 1$
Explicit $r(z)$	Varies (e.g., $\frac{1}{4z^2}$, $\frac{-1+3z^2}{4z^2(z^2-1)^2}$)	$\frac{3}{16\,z^2(z-1)^2}$
Algebraic extension	$\mathbb{C}(z, \sqrt{az+b})$ (linear radicand)	$\mathbb{C}(z, \sqrt{1/(z(z-1))})$
Construction direction	Given $r(z)$, find $s(z)$	Given $s(z)$ structure, derive $r(z)$
Residue pattern	Various	Equal residues $\frac{1}{4}$ at each pole

In the examples from Kovacic and the references we consulted [1, 2, 3, 4], we did not find this exact inverse two-pole construction. The construction is natural from the perspective of inverse Riccati problems but appears to be less common in the examples we consulted.

9 Conclusion

We have presented a systematic construction of second-order linear ODEs with degree-2 Liouvillian solutions, based on a two-pole ansatz for the logarithmic derivative $s(z) = A(z) + B(z)\sqrt{R(z)}$. The specific example with poles at $z = 0$ and $z = 1$ yields the ODE

$$y'' + \frac{3}{16\,z^2(z-1)^2}\,y = 0$$

with solutions:

$$y_1 = \big[z(z-1)\big]^{1/4}\,\big[2z - 1 + 2\sqrt{z(z-1)}\big]^{1/2},$$
$$y_2 = \big[z(z-1)\big]^{1/4}\,\big[2z - 1 - 2\sqrt{z(z-1)}\big]^{1/2}.$$

The construction was verified by: (i) direct computation of $r(z)$ from $A' + A^2 + B^2R$, (ii) proof that s has algebraic degree exactly 2 via its minimal polynomial, (iii) proof of linear independence via the nonconstant ratio $y_1/y_2 = u$ and constant nonzero Wronskian $W = \pm 1$, and (iv) supporting SymPy computational checks.

The n-pole generalization extends the framework to arbitrary numbers of poles with odd residue parameters, with the simplest subfamily ($f(z) = 1$) yielding constant $B = c$. Further exploration of non-constant $f(z)$ and the resulting ODE families is a natural direction for future work.

References

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